Methods Practice SAC3

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58 -

CAS + No CAS | 10m Reading | 1h 40m Writing | ~~59~~ marks

*All questions are No CAS except for those labelled with '**CAS ACTIVE**'.*
Please only use the CAS for these questions.

Question 1 (7 marks)

A student takes a multiple choice exam in which each question has 5 options, exactly one of which is correct.

If the student knows the correct answer, then they select that option. Otherwise, they just randomly select one of the 5 options.

Suppose the student knows the answer to 50% of the questions.

For any given question, we define the events C and K as:

C: the student picked the correct option.

K: the student knew the answer.

a. Determine each of: (2 marks)

$Pr(K)$ $Pr(C \mid K)$ $Pr(C \mid K')$

$Pr(K) = \frac{1}{2}$ $Pr(c|K) = 1$ $Pr(c|K') = \frac{1}{5}$

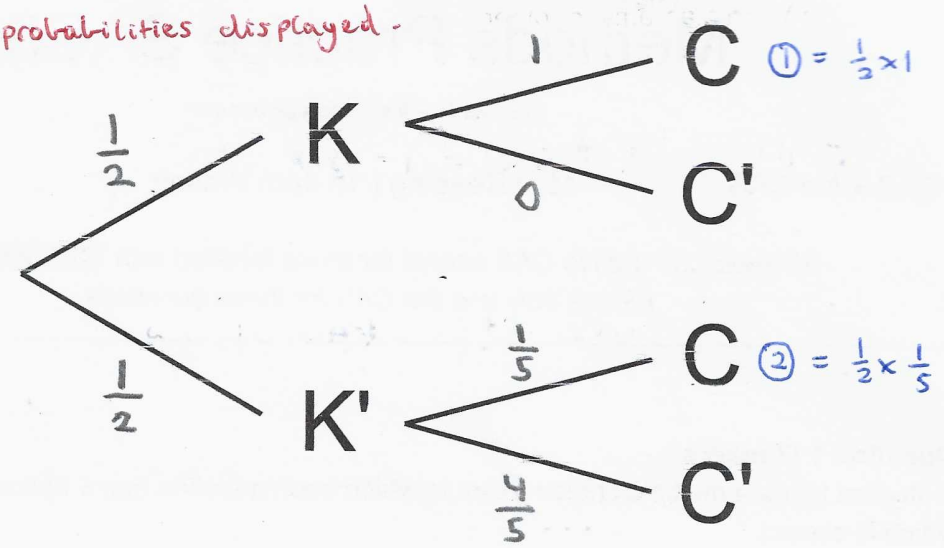
given. they know the answer, so they will 100% get it right. they're randomly guessing.

1M: at least one is right.

1M: all are right.

b. Model these probabilities, and their complements $Pr(K')$, $Pr(C' \mid K)$ and $Pr(C' \mid K')$, on the tree diagram below: (1 mark)

1M: all 6 probabilities displayed.



c. Hence, or otherwise, find the probability that the student picked the correct option. That is, find $Pr(C)$. (2 marks)

Law of total probability:

$Pr(C) = Pr(C|K)^{①} Pr(K) + Pr(C|K')^{②} Pr(K')$

$= (1)(\frac{1}{2}) + (\frac{1}{5})(\frac{1}{2})$ 1M

$= \frac{1}{2} + \frac{1}{10} = \frac{3}{5}$ 1A

d. Given the student picked the correct option, what is the probability that they actually knew the answer? (2 marks)

$Pr(K|C) = \frac{Pr(K \cap C)^{①}}{Pr(C)}$ 1M

$= \frac{1/2}{3/5} = \frac{5}{6}$ 1A

Question 2 (7 marks)

The events A and B satisfy the following equations:

$$Pr(A) = \frac{1}{3} \qquad Pr(A \cap B) = \frac{1}{4} \qquad Pr(A' \cap B) = \frac{1}{4}$$

With the help of Venn diagrams, or otherwise, find:

a. $P(A')$ (1 mark)

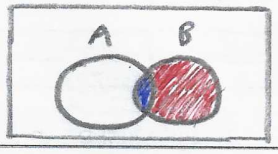
$Pr(A') = 1 - Pr(A) = 1 - \frac{1}{3} = \frac{2}{3}$ 1A

b. $P(B \mid A)$ (2 marks)

1M

$Pr(B|A) = \frac{Pr(B \cap A)}{Pr(A)} = \frac{1/4}{1/3} = \frac{3}{4}$ 1A

c. $P(B)$ (2 marks)

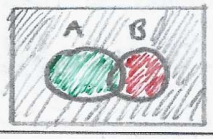


1M

$Pr(B) = Pr(A \cap B) + Pr(A' \cap B) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$ 1A

d. $P(A' \mid B')$ (2 marks)

$Pr(A'|B') = \frac{Pr(A' \cap B')}{Pr(B')}$



$= \frac{1 - Pr(A) - Pr(A' \cap B)}{1 - Pr(B)}$ 1M

$= \frac{1 - \frac{1}{3} - \frac{1}{4}}{1 - \frac{1}{2}} = \frac{\frac{5}{12}}{\frac{1}{2}} = \frac{5}{6}$ 1A

Question 3 (8 marks)

Neap 2020 Exam 1 Q6

this is without replacement!!
↑ use combinations! (probably).

A phone dealer has eight new J-phones in stock. It is found that three of these phones have faulty batteries.

a. Two of the eight J-phones are selected at random.

i. Find the probability that at least one of the selected phones has a faulty battery. 2 marks

Let F be the number of phones with a faulty battery.

$\rightarrow Pr(F \geq 1) = 1 - Pr(F=0) = 1 - \frac{\binom{3}{0}\binom{5}{2}}{\binom{8}{2}} = \frac{9}{14}$ 1M 1A

OR $= 1 - \frac{5}{8} \times \frac{4}{7} = \frac{9}{14}$ also 1M.

ii. Given that at least one of the selected phones has a faulty battery, find the probability that exactly one phone has a faulty battery. 2 marks

$Pr(F=1 \mid F \geq 1) = \frac{Pr(F=1 \cap F \geq 1)}{Pr(F \geq 1)} = \frac{Pr(F=1)}{Pr(F \geq 1)}$ ①

① $= \frac{\binom{3}{1}\binom{5}{1}}{\binom{8}{2}} = \frac{3 \times 5}{\frac{8!}{2!6!}} = \frac{15}{\frac{7 \times 8}{2}} = \frac{15}{28}$

$\Rightarrow pr = \frac{15/28}{9/14} = \frac{15}{28} \times \frac{14}{9} = \frac{15}{18} = \frac{5}{6}$ 1M 1A

b. It is known that, overall, 20% of J-phones will have a faulty battery. The probability of any given phone having a faulty battery is independent to any other phone. A phone dealer receives a delivery of four new J-phones.

Find the probability that at least two of these phones will have faulty batteries. 2 marks

Let $X \sim Bi(4, 1/5)$

$\rightarrow Pr(X \geq 2) = 1 - P(X=0) - Pr(X=1)$

$= 1 - \binom{4}{0}\left(\frac{1}{5}\right)^0\left(\frac{4}{5}\right)^4 - \binom{4}{1}\left(\frac{1}{5}\right)^1\left(\frac{4}{5}\right)^3$ 1M

$= 1 - \left(\frac{4}{5}\right)^4 - 4\left(\frac{4^3}{5^4}\right) = 1 - \left(\frac{4}{5}\right)^4 - \left(\frac{4}{5}\right)^4$

$= 1 - 2\left(\frac{4}{5}\right)^4 = 1 - 2\left(\frac{256}{625}\right) = 1 - \frac{512}{625}$

$= \frac{113}{625}$ 1A

alternatively, simply notice that b must be 1.5 standard deviations above the mean. $\Rightarrow b = 540 + 70 \times 1.5$

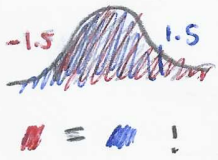
c. Working batteries have a lifespan, measured in minutes, after which they require recharging.

The lifespan of working batteries is a normally distributed variable, X, with a mean of 540 minutes and a standard deviation of 70 minutes. Let Z be the standard normal variable such that $Z \sim N(0, 1)$.

$X \sim \text{Norm}(540, 70^2)$

$\Rightarrow X = 70Z + 540$

i. Find b such that $\Pr(X < b) = \Pr(Z > -1.5)$. 1 mark



= !

$\text{LHS} = \Pr(70z + 540 < b) = \Pr\left(z < \frac{b-540}{70}\right)$

$\text{RHS} = \Pr(z < 1.5)$

$\Rightarrow \frac{b-540}{70} = 1.5 \iff b - 540 = 105$

$\Rightarrow b = 645$ 1A

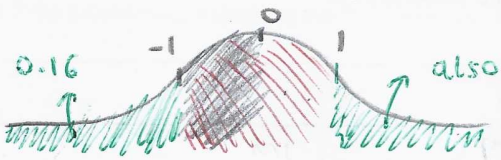
ii. Correct to two decimal places, $\Pr(Z < -1) = 0.16$.

Find the probability that a randomly selected phone battery has a lifespan of greater than 470 minutes, given that it has a lifespan of less than 540 minutes. 1 mark

$\Pr(X > 470 \mid X < 540) = \frac{\Pr(470 < X < 540)}{\Pr(X < 540)}$.

$= \frac{\Pr(-1 < z < 0)}{\Pr(z < 0)}$ ① ② $= \frac{0.34}{1/2} = 0.68$. 1A

① :



$\Rightarrow$ + 0.16 + 0.16 = 1

$\Rightarrow$ = 0.68

$\Rightarrow$ = 0.34 !

OR $\rightarrow$ + 0.16 = 0.5

$\Rightarrow$ = 0.34

② :



$\downarrow$ clearly $\frac{1}{2}$ lol.

Question 4 (2 marks)

Neap 2021 Exam 1 Q6

An animal rescue shelter has a population of rabbits of different breeds. European rabbits form $\frac{3}{5}$ of the population. If a sample of size n is taken from the population, $\hat{P}$ represents the proportion of European rabbits in the sample.

Find the value of n such that the standard deviation of $\hat{P}$ is equal to $\frac{1}{10}$.

In general, $sd(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$

$\rightarrow \sqrt{\frac{(\frac{3}{5})(\frac{2}{5})}{n}} = \frac{1}{10}$ 1M

$\Leftrightarrow \frac{6/25}{n} = \frac{1}{100}$

$\Leftrightarrow 100 \times \frac{6}{25} = n$

$\Rightarrow n = 24$. 1A

Question 5 (3 marks)

Neap 2021 Exam 1 Q5

The results of four independent trials show that $4\Pr(X=1) = \Pr(X=3)$, where the random variable X represents the number of successes. X is binomially distributed.

If p is the probability of success on any trial, find the value(s) of p.

$X \sim Bi(4, p)$.

$\rightarrow 4\Pr(x=1) = \Pr(x=3)$

$\Leftrightarrow 4\binom{4}{1}p(1-p)^3 = \binom{4}{3}p^3(1-p)$ 1M

$\Leftrightarrow 4p(1-p)^3 = p^3(1-p) \Leftrightarrow 4p(1-p)^3 - p^3(1-p) = 0$

$\Leftrightarrow p(1-p)\left(4(1-p)^2 - p^2\right) = 0$

$4 - 8p + 4p^2 - p^2 = 3p^2 - 8p + 4$.

$\Leftrightarrow p(1-p)(3p^2 - 8p + 4) = 0$ 1M

$p = 0 \quad p = 1$

$p = \frac{8 \pm \sqrt{64 - 48}}{6} \nearrow 16$

$= \frac{8 \pm 4}{6} = \frac{4 \pm 2}{3}$.

$\rightarrow p = \frac{2}{3}$ or ~~$p = 2$~~ ↗ not allowed!

$p \in [0, 1]$.

1A

$\therefore p = 0, \frac{2}{3}, 1$.

Question 6 (7 marks)

Neap 2019 Exam 1 Q6

Let X be a continuous random variable with the probability density function

$$f(x)=\begin{cases} kx^2(1-x) & 0\le x\le 1 \\ 0 & \text{elsewhere} \end{cases}$$

a. Show that $k = 12$. 2 marks

Require: $\int_0^1 Kx^2(1-x)\,dx = 1$ 1M

$\rightarrow$ LHS $= K\int_0^1 x^2 - x^3\,dx = K\left[\frac{1}{3}x^3 - \frac{1}{4}x^4\right]_0^1$

$= K\left(\frac{1}{3}-\frac{1}{4}\right) = \frac{K}{12}$

1A

$\rightarrow \frac{K}{12} = 1 \Rightarrow K = 12$.

b. Find the mean of X. 2 marks

$E(X) = \int_0^1 x\left(Kx^2(1-x)\right)dx = K\int_0^1 x^3 - x^4\,dx$ 1M

$= K\left[\frac{1}{4}x^4 - \frac{1}{5}x^5\right]_0^1 = K\left(\frac{1}{4}-\frac{1}{5}\right) = \frac{K}{20}$

$= \frac{12}{20} = \frac{3}{5}$ 1A

c. Let Y be a continuous random variable with the probability density function

$$g(x)=\begin{cases} qx^2(p-x) & 0\le x\le p \\ 0 & \text{elsewhere} \end{cases}$$

Given that $p, q > 0$ and the mean of Y is equal to $\frac{3}{10}$, find the value of p and q. 3 marks

Require: $\begin{cases} \int_0^p qx^2(p-x)\,dx = 1 \ \text{①} & (\text{area} = 1) \\ \int_0^p qx^3(p-x)\,dx = \frac{3}{10} \ \text{②} & (E(X) = \frac{3}{10}) \end{cases}$ 1M

① : LHS $= q\left[\frac{1}{3}px^3 - \frac{1}{4}x^4\right]_0^p = q\left(\frac{1}{3}p^4 - \frac{1}{4}p^4\right) = \frac{qp^4}{12}$

② : LHS $= q\left[\frac{1}{4}px^4 - \frac{1}{5}x^5\right]_0^p = q\left(\frac{1}{4}p^5 - \frac{1}{5}p^5\right) = \frac{qp^5}{20}$

$\Rightarrow \begin{cases} qp^4 = 12 \ \text{ⓐ} \\ qp^5 = 6 \ \text{ⓑ} \end{cases}$ 1M

$\rightarrow \frac{\text{ⓑ}}{\text{ⓐ}} \Leftrightarrow p = \frac{6}{12} = \frac{1}{2}$.

ⓐ $\Leftrightarrow q\left(\frac{1}{16}\right) = 12$ 1A

$\Rightarrow q = 192$.

12×16

12
60
20
100
192

Question 7 (11 marks) **CAS ACTIVE**

Neap 2019 Exam 2 Q2

The number of hamburgers sold at a fast-food restaurant on any given day is normally distributed with a mean of 120 and a standard deviation of 15.

a. Find the probability that on any given day more than 150 hamburgers are sold. Give your answer correct to four decimal places. 1 mark

Let $H \sim Norm(120, 15^2)$

$\rightarrow Pr(H > 150) = 0.0228$. 1A

b. If the restaurant sells at least x hamburgers on at least 75% of days, find the value of x to the nearest whole number. 1 mark

$Pr(X > x) = 0.75$



$\Leftrightarrow Pr(X < x) = 0.25$ → Use invnorm $\Rightarrow x = 110$.

1A

It is known that 70% of customers select a hamburger with cheese. A random sample of 100 customers was taken.

c. **i.** What is the probability, correct to four decimal places, that more than 75% of customers selected a hamburger with cheese? Do **not** use a normal approximation. 2 marks

Let $C \sim Bi(100, 0.7)$ (C is no. that have cheese) .

→ more ↕ than 75% of 100 = 76 or more!

→ $Pr(C \geq 76) = 0.1136$.

1M 1A

ii. What is the answer to **part c.i.** if a normal approximation is used? Give your answer correct to four decimal places. 2 marks

or say $\hat{P} \sim Norm(0.7, (\frac{\sqrt{21}}{100})^2)$

Find $Pr(\hat{P} > 0.75)$.

Binomial RVs can be approximated with normal distributions!

→ $E(C) = 70$, $Var(C) = 70 \times 0.3 = 21$

⇒ Let $C_{norm} \sim Norm(70, 21)$ 1M

→ $Pr(C_{norm} > 75) = 0.1376$ 1A

In order to estimate the proportion of customers who like lettuce in their hamburger, another random sample was taken.

It found that out of 500 customers, 200 of them like lettuce in their hamburger.

d. Find a 95% confidence interval for the proportion of customers that like lettuce in their hamburger. (1 mark)

$\hat{P} = \frac{200}{500} = 0.4$ → conf. int. = (0.3571, 0.4429) 1A

↓ or just use '1-Prop z Interval'.

wider!

check via CAS! 99% → (0.3436, 0.4564).

e. Determine whether a 99% confidence interval would be narrower or wider than a 95% confidence interval. Explain why. (2 marks)

Wider. 1A

A 99% is more confident that the population proportion p lies within it, meaning it needs to be wider to cover more values. 1M

Another sample of n customers found that the sample proportion of those who like lettuce in their hamburger was $\hat{p} = 0.45$.

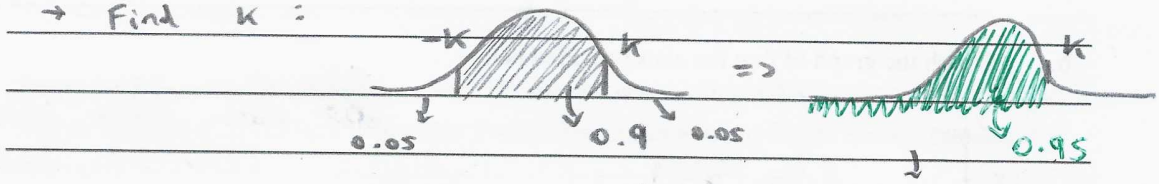
A 90% confidence interval was then constructed, accurate to 4 decimal places:

$$(0.4211, 0.4789)$$

f. Find the sample size n that resulted in this confidence interval. (2 marks)

Conf. int. $= \left(\hat{p} - k\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}, \ldots\right)$

→ Find k:



invnorm!

$\Rightarrow k = 1.6449\ldots$

$\rightarrow 0.45 - 1.6449\sqrt{\frac{(0.45)(0.55)}{n}} = 0.4211$ 1M

$\Rightarrow n = 801.74\ldots$

$\therefore$ Sample size $n = 802$! 1A

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Question 8 (~~14~~ marks) **CAS ACTIVE**

Neap 2019 Exam 2 Q4

Max is learning to juggle. The time that he can juggle before dropping a ball, t, in minutes, can be modelled by the probability density function $f(t)$.

$$f(t) = \begin{cases} t & 0 \le t \le \frac{2}{3} \quad F_1(t) \\ k(t-3) & \frac{2}{3} < t \le 3 \quad F_2(t) \\ 0 & \text{elsewhere} \end{cases}$$

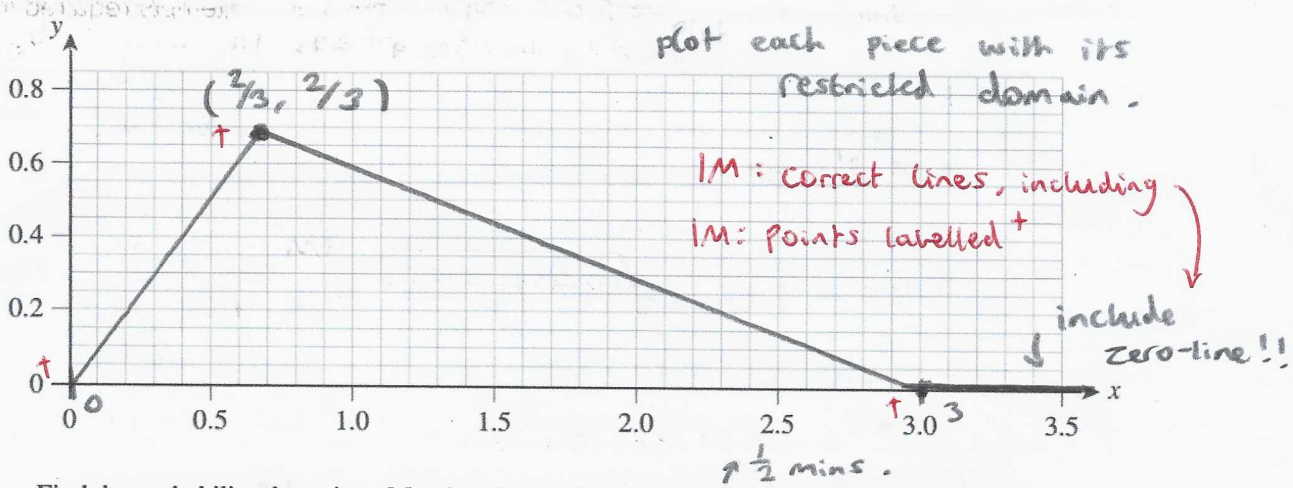
a. Show that $k = -\frac{2}{7}$. 3 marks

Require $\int_0^3 F(t)\,dt = 1$ 1M

$\rightarrow$ LHS $= \int_0^{2/3} \underbrace{t}_{F_1(t)}\,dt + \int_{2/3}^3 \underbrace{k(t-3)}_{F_2(t)}\,dt$ 1M

$= \frac{2}{9} - \frac{49k}{18} + \frac{2}{9} - \frac{49k}{18} = 1 \Rightarrow k = -\frac{2}{7}$. 1M

b. Sketch the graph of f on the axes provided below. 2 marks



plot each piece with its restricted domain.

1M: correct lines, including

1M: points labelled

include zero-line!!

$\frac{1}{2}$ mins.

c. Find the probability that, given Max juggles for longer than 30 seconds, he juggles for at least 2 minutes. 3 marks

Let T be the time Max juggles for.

$\rightarrow \Pr(T > 2 \mid T > \frac{1}{2}) = \frac{\Pr(T > 2)}{\Pr(T > \frac{1}{2})}$ 1M

$= \frac{\int_2^3 F_2(t)\,dt}{\int_{1/2}^{2/3} F_1(t)\,dt + \int_{2/3}^3 F_2(t)\,dt} = \frac{1/7}{7/8} = \frac{8}{49}$ 1A

1M

$\int_a^3 f_2(t)\,dt = 0.6 \Rightarrow a = -\frac{\sqrt{105}-15}{5}$ 1A

↳ where $a \in [2/3, 3]$

d. Find a such that $\Pr(T > a) = 0.6$. 2 marks

$\int_a^3 f(t)\,dt = 0.6$ → Check second piece's area: $\int_{2/3}^3 f_2(t)\,dt = \frac{7}{9} \approx 0.77 > 0.6$ 1M

⇒ a must be in the 2nd piece, $a \in [\frac{2}{3}, 3]$.

→ Now solve †

Max deems an individual attempt where he juggles for at least two minutes as "successful'.

Assume that the amount of time that Max juggles in any given attempt is independent of any other attempt.

e. Find the probability that any given attempt is 'successful'. 1 mark. ~~(2 marks)~~

$\Pr(\text{successful}) = \Pr(T > 2) = \frac{1}{7}$ 1A

↳ from (c).

↳ $\int_2^3 f_2(t)\,dt$

f. Using $invBinomN$, or otherwise, find the minimum number of attempts required for Max to ensure that he has a greater than 70% chance of **at least three** 'successful' attempts. (2 marks)

Let $S \sim Bi(n, \frac{1}{7})$ 1M (S = no. successful attempts).

→ Require: $\Pr(S \geq 3) > 0.7$

<=> $\Pr(S \leq 2) < 0.3$

→ Plug into invBinomN

⇒ $n = 25$ 1A.